The Learnability of Model-Theoretic Interpretation Functions in Neural Networks

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Zoom link: Adrian available to discuss during poster session (1:40-3:20pm, local time)

Central Question

Can NNs learn model-theoretic interpretation functions that generalize systematically to novel sentences?

The Systematicity Challenge

Fodor & Pylyshyn (1988) argued: Human cognition is **systematic** (understanding "cat on mat" implies understanding "dog on mat"); Systematicity requires compositional, symbolic representations; NNs systematic only by implementing a symbol system

Frank et al. (2009): Networks can exhibit systematicity **without** implementing symbols; Systematicity arises from **structure in the world**, encoded in analogical situation-vector representations; Situation vectors are NOT compositional/symbolic—no parts represent concepts

We extend this work with:

- Pool MCMC sampling (correct joint distribution)
- Entity vectors (semantic type e extension)
- Modern architectures (LSTM, Attention)
- Systematic competing event generation, which enables an extended eval set ($\approx 350-400$ vs. ≈ 100 sentences)

The Microworld

17 entities in 6 categories: **People** (3): charlie, heidi, sophia; **Games** (3): chess, hide&seek, soccer; **Toys** (3): puzzle, ball, doll; **Locations** (4): bathroom, bedroom, playground, street; **Manners** (4): well, badly, with_ease, with_difficulty

44 atomic propositions: play(person, game): 9; play(person, toy): 9; win(person), lose(person): 6; location(person, place): 12; play_manner(person, manner): 6; win_manner(manner): 2

173 hard constraints across multiple categories: Game rules: one game at a time, chess=2 players & soccer=2–3; Location rules: everyone has exactly one, chess players co-located; Winning rules: can't both win and lose, only one winner; Toy rules: puzzle=1 player only, soccer requires ball

Situation space: $2^{44} \approx 17.6$ trillion → **28,676** valid situations

The Microlanguage

40 words in a feature-based CFG: Persons, games, toys, locations; Verbs: plays, wins, beats, loses, is, played, won, lost; Modifiers: well, badly, inside, outside, with ease/difficulty; Prepositions: with, to, at, in, by

4 synonym pairs: charlie=boy, soccer=football, puzzle=jigsaw, bathroom=shower

Total: 13,556 sentences (only some consistent with hard constraints)

Example translations: "charlie plays chess" → play(charlie, chess); "girl plays chess" → play(heidi, chess) ∨ play(sophia, chess); "charlie loses to sophia" → win(sophia) ∧ lose(charlie)

Our Extensions to Frank et al. (2009)

I. Pool MCMC Sampling

Problem: Frank et al. sampled propositions *sequentially*, evaluating soft constraints on partial information. This underestimates correlations.

Our solution:

1. Pre-enumerate all 28,676 valid situations
2. MCMC over pool with Metropolis-Hastings
3. Acceptance based on **full-context** soft constraint probability
4. 1M raw samples → 250k thinned (factor 4)

Result: Correct joint distribution respecting both hard AND soft constraints.

II. Entity Vectors

Motivation: Formal semantics has two fundamental types—truth values (type t) AND entities (type e). Frank et al. encoded only truth values.

Implementation:

1. Enhance MCMC samples: 44-dim → 61-dim (+ 17 entity indicators)
2. Joint competitive layer training on 61-dim samples
3. Extract: 150×44 props + 150×17 entities

Training targets: Without entities: 150-dim; With entities: 300-dim (concatenated)

III. Modern Architectures

Capacity-matched at ≈66k parameters:

Arch	Hidden	Layers	Heads	Params
SRN	178	1	—	66,010
LSTM	80	1	—	66,950
Attn AbsPE	48	2	4	65,670
Attn RoPE	48	2	4	65,670

AbsPE: Sinusoidal positional encoding (Vaswani et al. 2017)
RoPE: Rotary positional encoding (Su et al. 2024)

IV. Principled Generation of Competing Events

Problem: Frank et al. used hardcoded rules per test group.

Our approach: Universal two-step competing event generation

Minimal difference for candidate generation: Vary exactly one event slot (agent, theme, location, manner) at a time.

Step 1 (Validity): Candidate + constraints → SAT
Step 2 (Competition): Candidate + described + constraints → UNSAT

Situation Vectors & Evaluation

Competitive Layer Training:

- Input: 250k thinned MCMC samples (44+17 = 61-dim)
- 150 competitive units, L1 distance, winner-take-all
- Output: 150×61 weight matrix $\mu \in [0, 1]$

Probability estimation (for any propositions a, b):

$$P(a) \approx \frac{\sum_i \mu_i(a)}{150} \quad P(a \wedge b) \approx \frac{\sum_i \mu_i(a) \cdot \mu_i(b)}{150} \quad P(a|b) = \frac{P(a \wedge b)}{P(b)}$$

Fuzzy logic operations (truth conditions for non-atomic props):

$$\mu_i(\neg a) = 1 - \mu_i(a)$$
$$\mu_i(a \wedge b) = \mu_i(a) \cdot \mu_i(b)$$
$$\mu_i(a \vee b) = \mu_i(a) + \mu_i(b) - \mu_i(a) \cdot \mu_i(b)$$

Comprehension Score (Frank et al. Eq. 7):

Measures attained fraction of maximum possible belief change:

$$\text{Comprehension}(a|z) = \begin{cases} \frac{P(a|z)-P(a)}{1-P(a)} & \text{if } P(a|z) > P(a) \\ \frac{P(a|z)-P(a)}{P(a)} & \text{otherwise} \end{cases}$$

Described vs. Competing Events:

Example: "charlie beats heidi" ⇒ win(charlie) ∧ lose(heidi)

- **Described**: win(charlie), lose(heidi)—should be **positive**
- **Competing**: win(heidi), lose(charlie)—should be **negative**

Systematicity = **Advantage** = score(descr) - score(compt) ⇒ model correctly derives specific target truth conditions

Train/Test Design

Complementary splits: What's held out in Split 1 is trained in Split 2 (and vice versa)—ensuring results don't depend on which particular sentences are excluded (C=charlie, H=heidi, S=sophia).

Test Group	Split 1 Held Out	Split 2 Held Out	Truth Cond in Train
Word (easiest)	C+soccer	boy+football	yes
Sentence	C {beats/loses to} H …	C {beats/loses to} S …	yes
Complex Event	chess+outside …	chess+inside …	no
Basic (hardest)	C+doll …	C+ball …	no

Four test groups of increasing difficulty:

- **Word**: Novel word combinations (synonym substitution), but target truth cond seen in training
- **Sentence**: Novel person pairs in "beats"/"loses to"—can model learn argument alternations?
- **Complex**: Novel game+location conjunctions, target truth cond **not** seen in training (only truth conditions for individual conjuncts seen)
- **Basic**: Novel person+toy combinations (hardest), target truth cond **not** seen in training

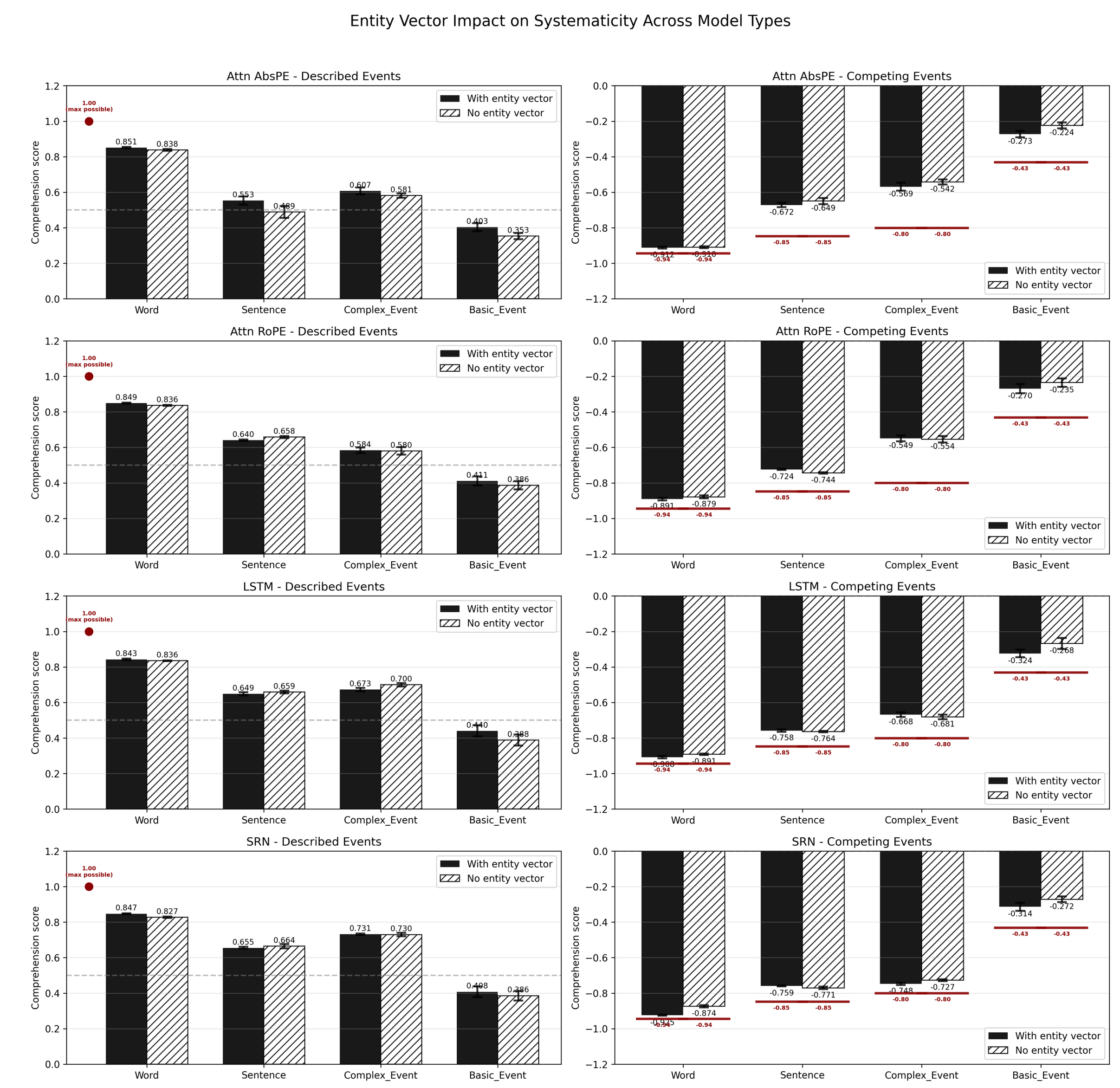
Grammar generates 13,556 sentences, split into train/test:

File	Total	Consistent	Toy	Non-Toy
train_set1	9,443	**6,279**	395 (6.3%)	5,884
train_set2	9,443	**6,789**	399 (5.9%)	6,390
test_set1	4,113	**2,840**	70 (2.5%)	2,770
test_set2	4,113	**2,330**	66 (2.8%)	2,264

Sentence **consistent** iff described event possible in at least one valid microworld situation. Only consistent sentences are used for training and evaluation. Repeat toy sent.s in training $4\times$ to give NNs best chance for basic event test.

Test-Only Described & Competing Scores

80 models total = 4 arch × 2 entity × 2 splits × 5 seeds



Left: described events. Right: competing events. Hatched: no entity vectors. Solid: with entity vectors. Dark red lines: ideal model performance.

Test-Only Advantage Scores

		Word	Sent	Cmplx	Basic
SRN	no ent	1.70	1.44	1.46	0.66
	w/ ent	1.77	1.42	1.48	**0.72**
LSTM	no ent	1.73	1.43	1.39	0.67
	w/ ent	1.75	1.42	1.35	**0.77**
Attn AbsPE	no ent	1.75	1.14	1.14	0.58
	w/ ent	1.76	1.24	1.19	0.68
Attn RoPE	no ent	1.71	1.41	1.15	0.63
	w/ ent	1.74	1.38	1.15	0.69

Conclusions

Main result: Entity vectors consistently aid novel entity combination generalization; Basic Event: +0.06 to +0.10 across all architectures

Open question: Why do recurrent architectures generalize better to novel basic events?

Next step: Match Attention capacity w/ ent to RNN/LSTM (Capacity of Attention w/ ent ≪ RNN/LSTM w/ ent)

[1] Frank, Haselager & van Rooij (2009). Connectionist semantic systematicity. *Cognition*, 110(3), 358–379. [2] Fodor & Pylyshyn (1988). Connectionism and cognitive architecture. *Cognition*, 28, 3–71. [3] Vaswani et al. (2017). Attention is all you need, *NeurIPS 31*, 1–15. [4] Su et al. (2021). RoFormer: Rotary position embedding. *Neurocomputing*, 450, 104–112. [5] Venhuizen et al. (2022). Distributional formal semantics. *Info. & Comp.*, 287, 1–21.